

16.Recipe Preparation Agent

Problem Statement No. : 16 –

Problem Statement No. 16- Recipe Preparation Agent The Challenge - A Recipe Preparation Agent helps users cook meals using only the ingredients they have on hand. By inputting available groceries, users receive tailored recipe suggestions using a RAG-based AI system. The agent retrieves relevant recipes and generates step-by-step instructions adapted to ingredient limitations. It offers substitutions, cooking tips, and dietary adjustments based on user preferences or restrictions. Designed to reduce food waste and save time, it turns pantry items into practical meal solutions. This AI assistant makes everyday cooking smarter, simpler, and more sustainable.

Objectives

- Recommend recipes from available ingredients.
- Reduce food waste through intelligent ingredient utilization.
- Provide nutrition and ingredient substitution suggestions.
- Demonstrate a multi-agent AI workflow using RAG concepts.

Technology Stack

IBM Granite, LangFlow, RAG, Streamlit, Python, Pandas

Modules

1. Recipe Assistant Agent
2. Recipe Knowledge Agent
3. Ingredient Substitution Agent
4. Meal Planner Agent

Future Scope

Image-based ingredient recognition, voice assistant integration, nutrition tracking, and smart grocery planning.